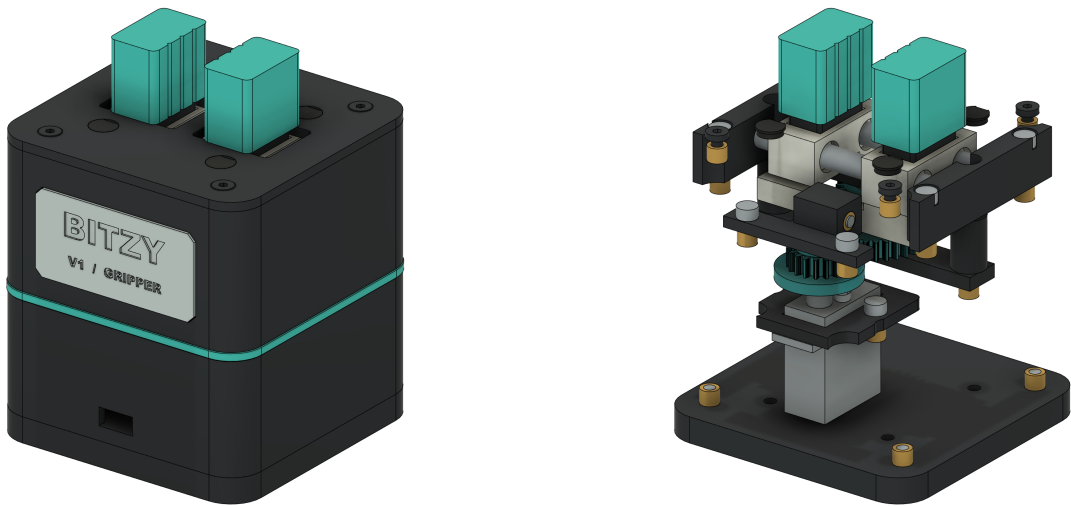


Parallel gripper / Rev B

A synchronized parallel-jaw mechanism with shared guide rods, replaceable TPU sleeves, removable access plugs and a serviceable PETG-CF enclosure.



Actual native CAD views of Gripper Rev B; the mechanism view reveals the gearing and guide arrangement.

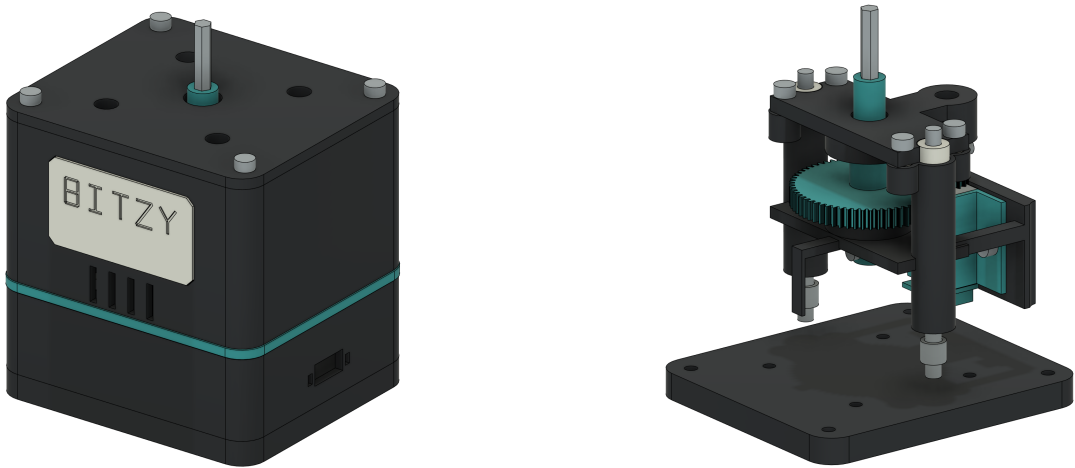
Nominal CAD reference	Value
Native model	108 parameters / 84 components
Soft-jaw opening / per-jaw travel	2-26 mm / 12 mm
Nominal case	74 x 73 mm
Printable set	25 product shapes + six fit coupons
Recorded review	Nine native poses, 181 gear-section positions; 31 meshes sliced

Physical grip force, printed clearances, friction, wear and whole-arm loads remain to be qualified. The original star and four mounting axes are retained. CAD/slicing success does not establish actual grip performance.

AI-assisted workflows supported the creation of this project.

Compliant screwdriver / R2

Encoder gearmotor, external reduction, replaceable printed bit socket and a supported cartridge with guided axial compliance. Completed digital release: September 9, 2026.



Actual native R2 CAD assembly and mechanism; physical fit, torque and durability remain unqualified.

Nominal CAD reference	Value
Native model	136 parameters / 67 components
Reduction / guided compliance	4:1 / 3 mm
Case / reference tip from mount	80 x 70 mm / 106 mm with nominal 28 mm bit
Installed mass	287.7 g estimate / 340 g planning allowance; not weighed
Printable set	12 product prints + 11 fit ladders
Recorded review	11 poses, six rotation transforms, 1,441 gear sections; 23 meshes sliced

Bit retention, runout, printed socket wear, motor current/encoder behavior, torque and duty need off-arm testing, followed by actual arm load/TCP qualification. No achieved output torque or complete-system strength result is claimed.